PQC DAILY STUDY SERIES

Real-World Latency: The Impact of PQC on TLS Handshake Times

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Introduction

As Post-Quantum Cryptography (PQC) standards like ML-KEM and ML-DSA are integrated into internet protocols, a critical question for network engineers and security architects is: "What is the performance impact?" The transition from classical algorithms (like ECDH and ECDSA) to their quantum-resistant counterparts is not a simple swap. PQC algorithms, while secure against quantum computers, often come with larger public keys, signatures, and ciphertexts. This directly impacts the amount of data exchanged during a cryptographic handshake, such as the one used in TLS (Transport Layer Security), which secures the vast majority of internet traffic (HTTPS).

Why PQC Increases TLS Handshake Latency

The TLS handshake is a sequence of messages exchanged between a client (e.g., your browser) and a server to establish a secure session. The latency of this handshake—the time it takes to complete—is a critical performance metric. Several factors related to PQC can increase this latency:

- **Larger Key Sizes:** PQC Key Encapsulation Mechanisms (KEMs) like ML-KEM have significantly larger public keys and ciphertexts compared to their classical counterpart, Elliptic Curve Diffie-Hellman (ECDH). For example, an X25519 public key is 32 bytes, whereas an ML-KEM-768 public key is 1184 bytes.
- **Increased Computational Load:** While often highly optimized, the mathematical operations in lattice-based cryptography (the foundation of ML-KEM and ML-DSA) can be more computationally intensive than elliptic curve operations, although this impact is often smaller than the size increase.
- **Network Packet Fragmentation:** The larger size of PQC keys and certificates can cause TLS handshake messages to exceed the standard Maximum Transmission Unit (MTU) of a network packet (~1500 bytes). This leads to IP fragmentation, where a single message is split into

multiple packets, requiring reassembly at the destination. Packet loss during fragmentation can lead to significant delays and connection failures.

Comparative Analysis: Handshake Latency Metrics

Let's compare a classical TLS 1.3 handshake with a hybrid one using both a classical (X25519) and a post-quantum (ML-KEM-768) key exchange. The hybrid approach is the most common deployment strategy, ensuring security against both classical and quantum adversaries.

Metric	Classical TLS (X25519)	Hybrid TLS (X25519 + ML-KEM-768)	Impact Analysis
ClientHello Size	~200-300 bytes	~1400-1500 bytes	The need to include the public key for ML-KEM drastically increases the size of the initial client message.
Server Hello + Certificate Size	~1-2 KB (variable)	~3-5 KB (with composite ML-DSA/RSA certs)	Post-quantum signatures in certificates (ML-DSA) are larger, increasing the size of server-side messages.
Round-Trip Time (RTT) Impact	Minimal	Can add 10-100ms or more, highly dependent on network conditions and packet loss.	Increased data means more packets, making the handshake more sensitive to network RTT and congestion.
Computational Time (CPU)	~0.1-1 ms	~0.5-5 ms	PQC operations are slightly slower, but this is often negligible compared to the network latency impact.

Real-World Applicability and Mitigation Strategies

The added latency is a significant concern for high-performance applications and users on low-quality networks (e.g., mobile or satellite). However, several strategies can mitigate this impact:

- **CDN and Edge Computing:** Terminating TLS connections closer to the user at an edge server reduces the Round-Trip Time (RTT), minimizing the effect of larger handshake messages.

- **TLS False Start & 0-RTT:** Session resumption mechanisms in TLS 1.3 (like 0-RTT) bypass the full handshake on subsequent connections, making the initial PQC overhead a one-time cost.
- **Algorithm Selection:** NIST Security Level 1 variants of PQC algorithms (e.g., ML-KEM-512) offer smaller sizes at the cost of a lower (but still very high) security margin, which may be acceptable for some use cases.

For most users on modern broadband connections, the added latency of a few dozen milliseconds for the initial connection will be imperceptible. However, for latency-sensitive services like real-time communication, financial transactions, or services for users in emerging markets, this overhead must be carefully measured and managed.

Daily Quiz

1. What is the primary reason PQC increases TLS handshake latency?
2. Why is IP fragmentation a concern for PQC in TLS?
3. Name one mitigation strategy to reduce the performance impact of PQC on web services.
